

Management Information Systems

Phase 3: Executive Business Intelligence Dashboards in Power BI

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1 Data Pipeline Cleansing & Custom DAX Architecture

1.1 Data Extraction, Transformation, and Loading (ETL)

During the preliminary audit of the raw dataset in the source Excel workbook, several structural anomalies were identified and rectified via the **Power Query (Transform Data)** pipeline:

- **Orphan Rows and Columns:** Redundant blank records and unindexed columns across individual sheets were pruned.
- **Type Coercion & Regional Formats:** Inconsistencies during schema inference regarding percentage fields and ISO calendar dates were standardized to ensure precise temporal aggregation.
- **Optimization of Quality Scoring:** The initial static formula for **Test_Score** within the Excel dataset exhibited sub-optimal sensitivity. Rather than patching the spreadsheet statically, the legacy column was purged, and a dynamic calculated column (**QCtestScore**) was implemented using DAX in Power BI to establish robust analytical relationships.

1.2 Formulation of Advanced DAX Calculations

1.2.1 1. Comminution Waste Percentage (Waste Percentage)

To quantify the operational reject rate across comminution runs, a calculated column was engineered on the **CrushingOperations** table, displayed on Page 1 of the dashboard:

$$\text{Waste Percentage} = \frac{\text{CrushingOperations}[\text{Waste(tons)}]}{\text{CrushingOperations}[\text{Input_Weight(tons)}]} \quad (1)$$

```
1 Waste Percentage = CrushingOperations[Waste(tons)] / CrushingOperations[Input_Weight(tons)]
```

Figure 1: DAX Calculated Column: Comminution Waste Percentage.

1.2.2 2. Composite Quality Assurance Metric (QCtestScore)

To evaluate the composite grade of assayed mineral specimens, a multi-parameter DAX calculated column was implemented. This metric credits copper purity while penalizing detrimental contaminants:

$$\begin{aligned} \text{QCtestScore} = & 0.6 \times \text{Purity_Percent} + 0.45 \times \text{Copper_Percent} - 0.1 \times \text{Iron_Percent} \\ & - 0.15 \times \text{Zinc_Percent} - 0.2 \times \text{Moisture_Percent} \end{aligned} \quad (2)$$

```
QCtestScore = 0.6*QualityControl[Purity_Percent]+0.45*QualityControl[Copper_Percent]-0.1*QualityControl[Iron_Percent]-0.15*QualityControl[Zinc_Percent]-0.2*QualityControl[Moisture_Percent]
```

Figure 2: DAX Calculated Column: Composite Quality Index (QCtestScore).

1.2.3 3. Fleet Monthly Logistics Expenditure (Monthly cost)

Formulated on the **Trucks** table and visualized on Dashboard Page 3 (Shipment), this measure determines the comprehensive operating cost per haul truck by synthesizing fuel consumption and scheduled/unscheduled maintenance liabilities:

- **Monthly Fuel Cost:** Average Daily Distance (km) × \$1.00/km × 30 operating days.
 - **Accident & Repair Liability:** Average Monthly Collisions × \$2,500 mean repair cost.
- $$\text{Monthly Cost} = (\text{AvgDistPerDay(km)} \times 1 \times 30) + (\text{AvgCarCrushPerMonth} \times 2500) \quad (3)$$

$$\text{monthly cost} = 1 * \text{Trucks}[\text{AvgDistPerDay}(\text{km})] * 30 + \text{Trucks}[\text{AvgCarCrushPerMonth}] * 2500$$

Figure 3: DAX Formulation: Monthly Logistics Operating Cost per Unit.

2 Dashboard Sheet 1: Waste & Warehouse Capacity Utilization

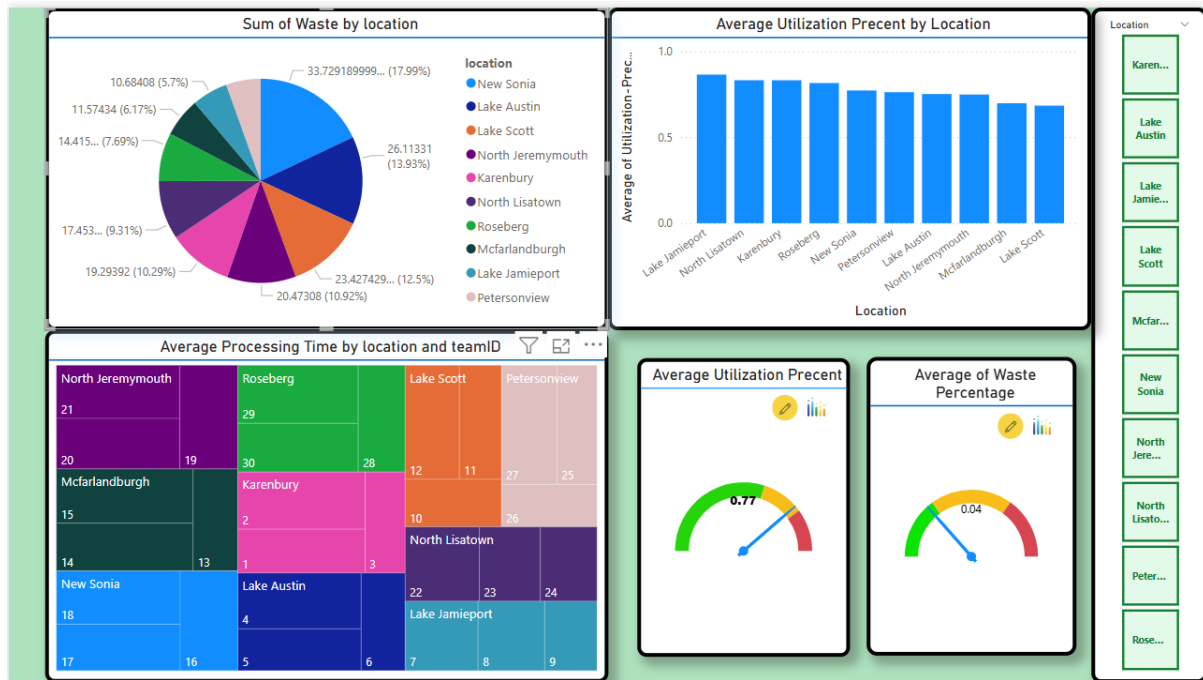


Figure 4: Dashboard Sheet 1 – Comminution Waste Generation and Regional Storage Utilization.

Sheet Architecture & Scope: Sheet 1 provides an executive-level diagnostic overview benchmarking comminution performance and warehouse inventory buffers across distinct geographic extraction nodes. A dedicated slicer positioned on the right allows dynamic filtering by municipal jurisdiction.

2.1 Visual Component Analysis

- 1. Sum of Waste by Location (Donut Chart):** Visualizes total cumulative waste tonnage segmented by operating region. The graphic clearly surfaces *New Sonia* as the largest contributor to comminution scrap, warranting immediate metallurgical and mechanical inspection.
- 2. Average Utilization Percent by Location (Clustered Column Chart):** Tracks warehouse inventory saturation relative to certified capacity. When a facility crosses critical capacity thresholds (e.g., 90%), it signals the necessity for emergency depot expansion or inventory re-routing.
- 3. Average Processing Time by Location and TeamID (Treemap Matrix):** Disaggregates mean comminution cycle durations across specific field squads. *Petersonview* achieves the fastest throughput, while leftward segments identify operational inefficiencies and training gaps.

2.2 Key Performance Indicators (KPIs)

KPI 1: Average Waste Percentage (Radial Gauge)

Quantifies the enterprise-wide reject percentage in comminution. An upward excursion beyond target operating bounds indicates excessive boulder hard-rock variance, worn jaw plates, or feeder miscalibrations, directly driving up unit operating costs.

KPI 2: Average Utilization Percent (Radial Gauge)

Reflects supply chain fluidity and dispatch velocity. Prolonged over-utilization highlights outbound bottlenecks, demonstrating that fleet haulage capacity must be expanded to maintain steady-state inventory drawdown.

3 Dashboard Sheet 2: Quality Assurance & Field Sampling Control

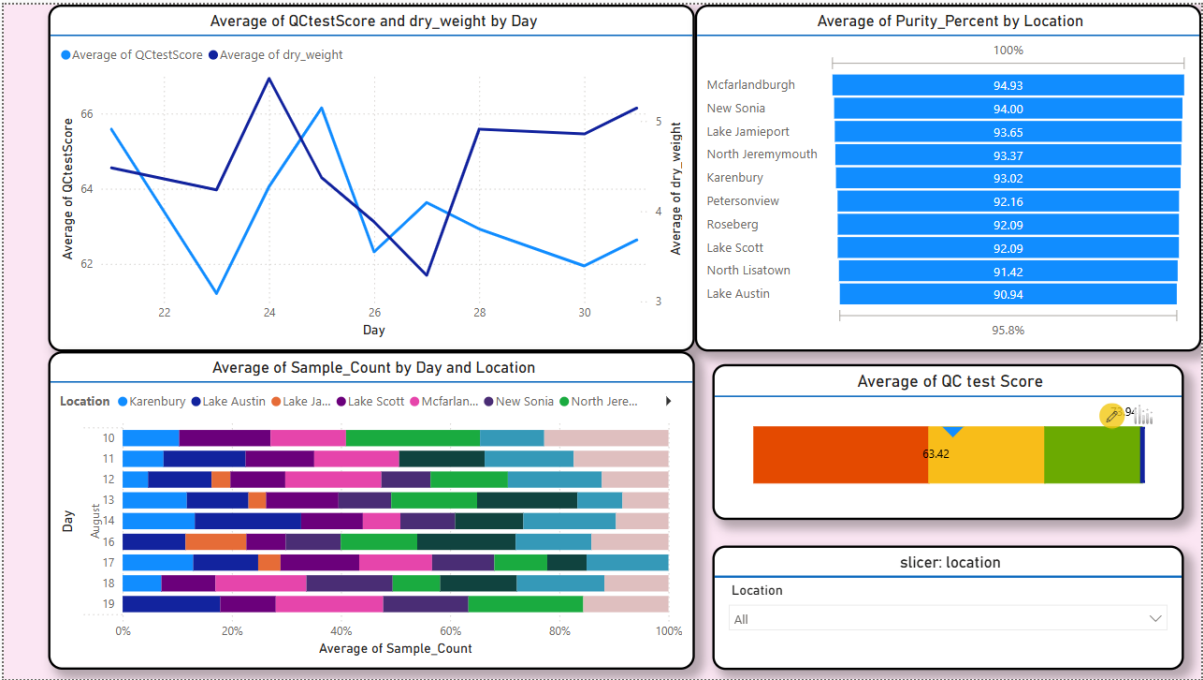


Figure 5: Dashboard Sheet 2 – Statistical Quality Control and Ore Grade Assays.

Sheet Architecture & Scope: Dedicated to metallurgical governance, Sheet 2 monitors specimen sampling discipline, moisture-to-grade dynamics, and elemental copper recovery across laboratory stations.

3.1 Visual Component Analysis

- 1. Average QCtestScore & Dry Weight by Day (Dual-Axis Line Chart):** Correlates daily dry sample mass against composite quality scores. The profiles demonstrate a strong positive correlation; however, an acute dip on days 26 and 27 flags anomalous low-grade throughput that requires investigation.
- 2. Average Sample Count by Day and Location (100% Stacked Bar Chart):** Audits sampling volume compliance across analytical centers. Significant declines (e.g., low collection volume at *Lake Jamieport*) pinpoint field crew negligence or equipment downtime.
- 3. Average Purity Percent by Location (Ranked Table/Bar):** Ranks primary mine sites by assay purity, directly guiding exploration drilling priorities toward highest-yield ore bodies.

3.2 Key Performance Indicator (KPI)

KPI 3: Average QC Test Score (Gauge & KPI Card)

The central technical benchmark measuring plant-wide metallurgical health. A sustained high score validates flotation and electrorefining precision, directly dictating product marketability and contract pricing tiers.

4 Dashboard Sheet 3: Fleet Logistics & Transportation Economics

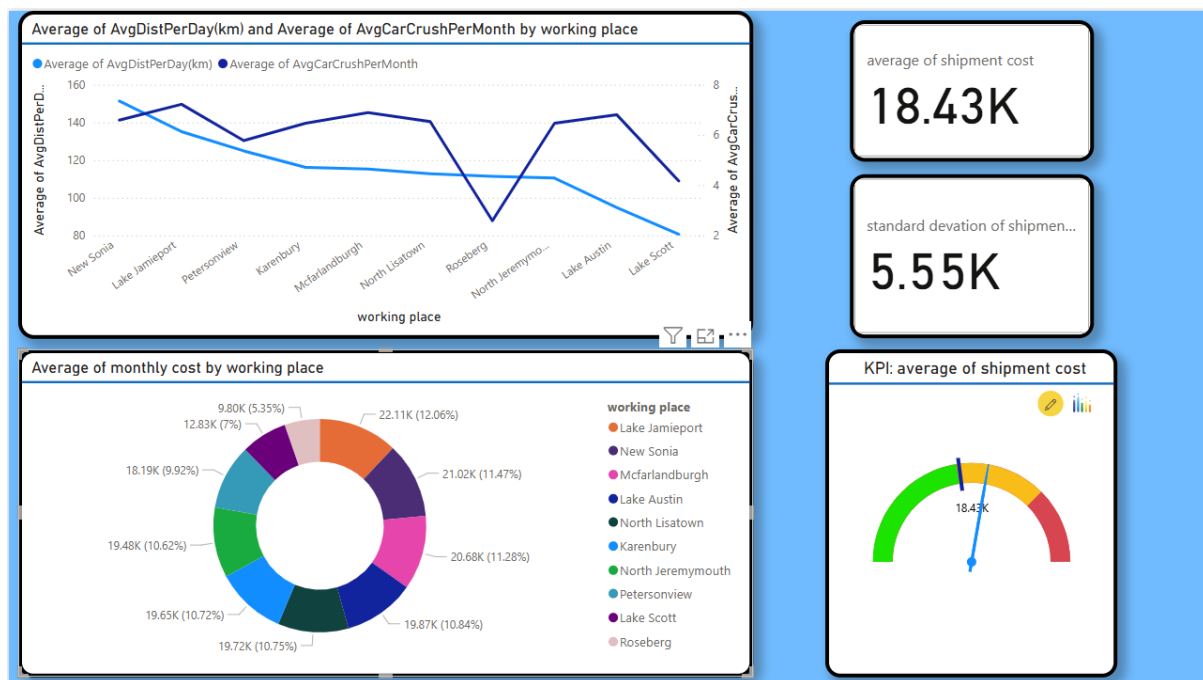


Figure 6: Dashboard Sheet 3 – Haul Fleet Safety, Distance Traveled, and Logistics Costs.

Sheet Architecture & Scope: Sheet 3 evaluates fleet utilization, transit safety hazards, and monthly expenditure across municipal transit routes.

4.1 Visual Component Analysis

- 1. Average Distance per Day vs. Monthly Accident Rate (Dual-Line Chart):** Correlates daily distance driven against accident frequency. Fleets demonstrating disproportionately high collision rates relative to mileage indicate hazardous haul-road geometry, necessitating route modifications.
- 2. Average Monthly Logistics Cost by Location (Donut Chart):** Disaggregates comprehensive hauling expenses per operating sector, directly supporting logistics route optimization.
- 3. Summary Metric Cards (Mean & Standard Deviation of Shipment Cost):** Provide instantaneous situational awareness regarding logistics burn rates. A spike in standard deviation signals irregular regional cost anomalies.

4.2 Key Performance Indicator (KPI)

KPI 4: Average Shipment Cost (Radial Gauge)

Tracks per-unit transportation expenses against budget ceilings. The gauge reflects an active warning condition exceeding target thresholds, mandating immediate route optimization and driver defensive training.

5 Dashboard Sheet 4: Energy Consumption & Carbon Footprint Analytics

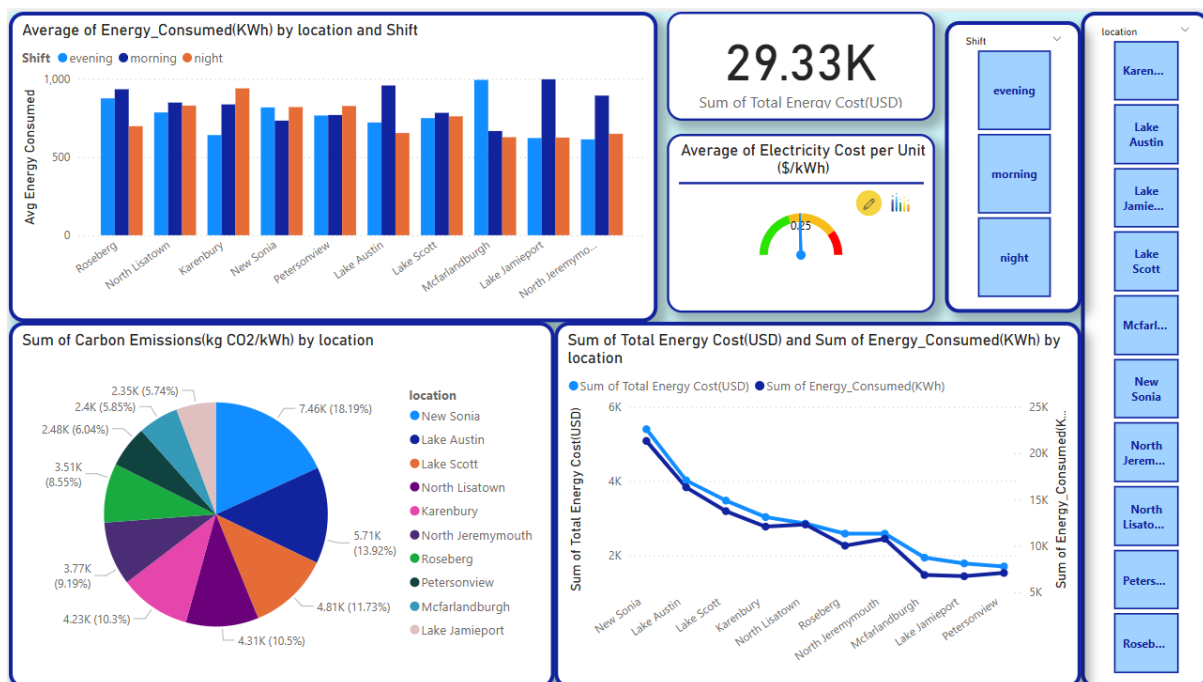


Figure 7: Dashboard Sheet 4 – Energy Consumption, Tariff Optimization, and Carbon Emissions.

Sheet Architecture & Scope: Sheet 4 provides deep visibility into electrical energy consumption, utility tariffs, and carbon footprints, supported by dual multi-select slicers (Shift and Location).

5.1 Visual Component Analysis

- 1. Average Energy Consumed (kWh) by Location and Shift (Clustered Column Chart):** Compares energy demand across shifts (Morning, Afternoon, Night), identifying peak-tariff inefficiencies.
- 2. Cumulative Carbon Emissions ($kg\ CO_2/kWh$) by Location (Donut Chart):** Quantifies greenhouse gas emissions across processing hubs to ensure compliance with corporate ESG mandates.
- 3. Total Energy Cost vs. Total Power Draw (Synchronized Dual Line Chart):** Continuously tracks the relationship between energy draw and billing cost. Divergence between these curves alerts management to adverse tariff adjustments or power factor penalties.

5.2 Key Performance Indicators (KPIs)

KPI 5: Cumulative Enterprise Energy Expenditure (Card Visualizer)

Monitors overall utility burn rate (\$29.33K baseline), providing immediate feedback on plant-wide efficiency initiatives.

KPI 6: Average Electricity Cost per Unit (\$/kWh) (Radial Gauge)

Tracks the effective electricity price per kilowatt-hour. This metric assesses whether operations are successfully shifted away from peak-hour tariff bands to off-peak night cycles.